What Drives Used Chevrolet Silverado Prices?

Regression modeling and diagnostic analysis of Southeast U.S. vehicle listings

Data Science Portfolio Project

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Executive summary

This analysis examines how **vehicle age** and **mileage** relate to advertised prices of 2010–2019 Chevrolet Silverado 1500 listings in the Southeast. The goal is to explain relationships, compare model complexity, and estimate out-of-sample predictive accuracy. Prices are *asking prices*, not verified transaction prices; associations are not causal.

Dataset

The provided source CSV is a used-vehicle listings extract. The reproducible portfolio dataset retains Silverado 1500 listings with model years 2010–2019, positive prices, and nonmissing nonnegative mileage. Age is measured relative to **2021**, matching the original coursework's reference year. The dataset includes 461 listings from 7 states. The analysis below uses this consistent subset for every model; the original homework sometimes used a different model label (`silverado`) and therefore different samples.

```
glimpse(vehicles)
```

```
## Rows: 461
## Columns: 5
## $ price    <dbl> 22590, 31990, 33995, 29800, 33990, 27897, 16477, 31990, 30900~
## $ year     <dbl> 2010, 2016, 2016, 2019, 2016, 2013, 2011, 2016, 2015, 2015, 2~
## $ odometer <dbl> 71229, 107597, 115000, 39000, 117895, 55033, 178349, 107597, ~
## $ state    <chr> "al", "al", "al", "al", "al", "al", "al", "al", "al", "al", "~
## $ age      <dbl> 11, 5, 5, 2, 5, 8, 10, 5, 6, 6, 5, 11, 10, 5, 10, 5, 11, 9, 8~
```

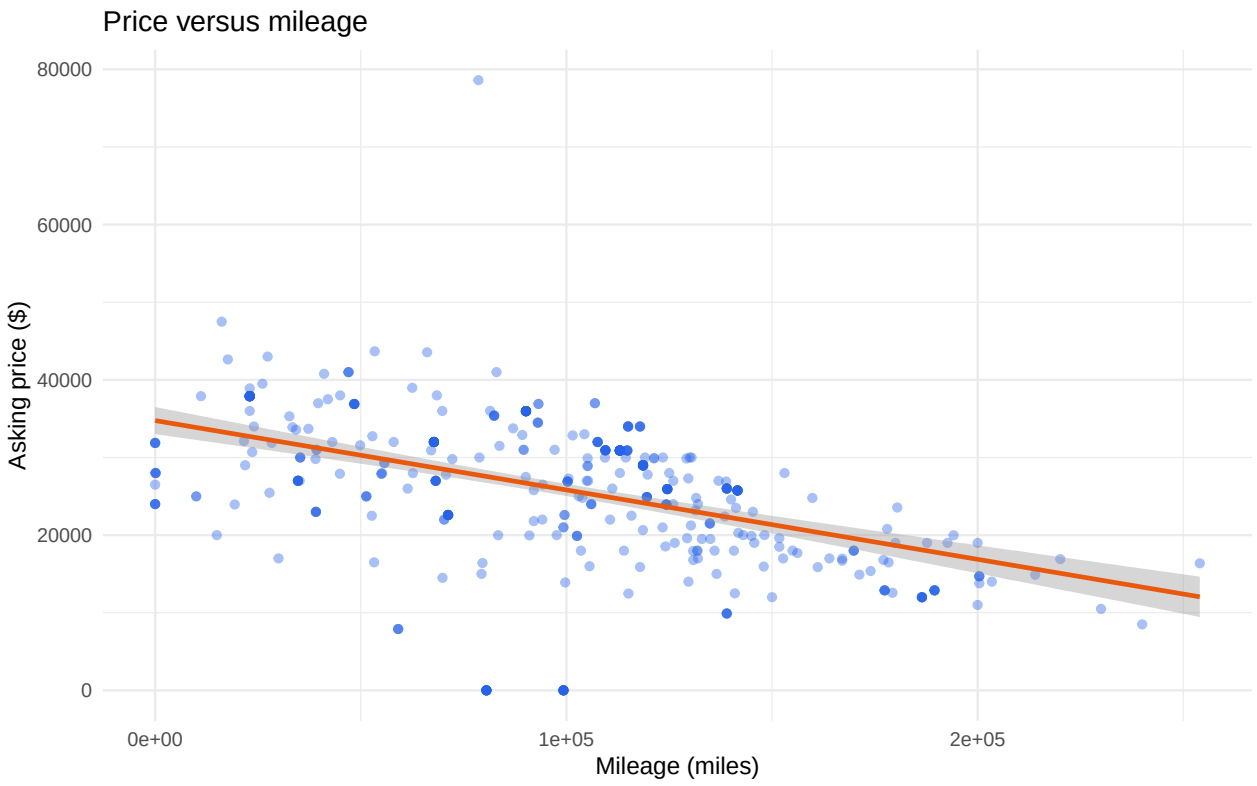
```
summary(vehicles[c("price", "age", "odometer")])
```

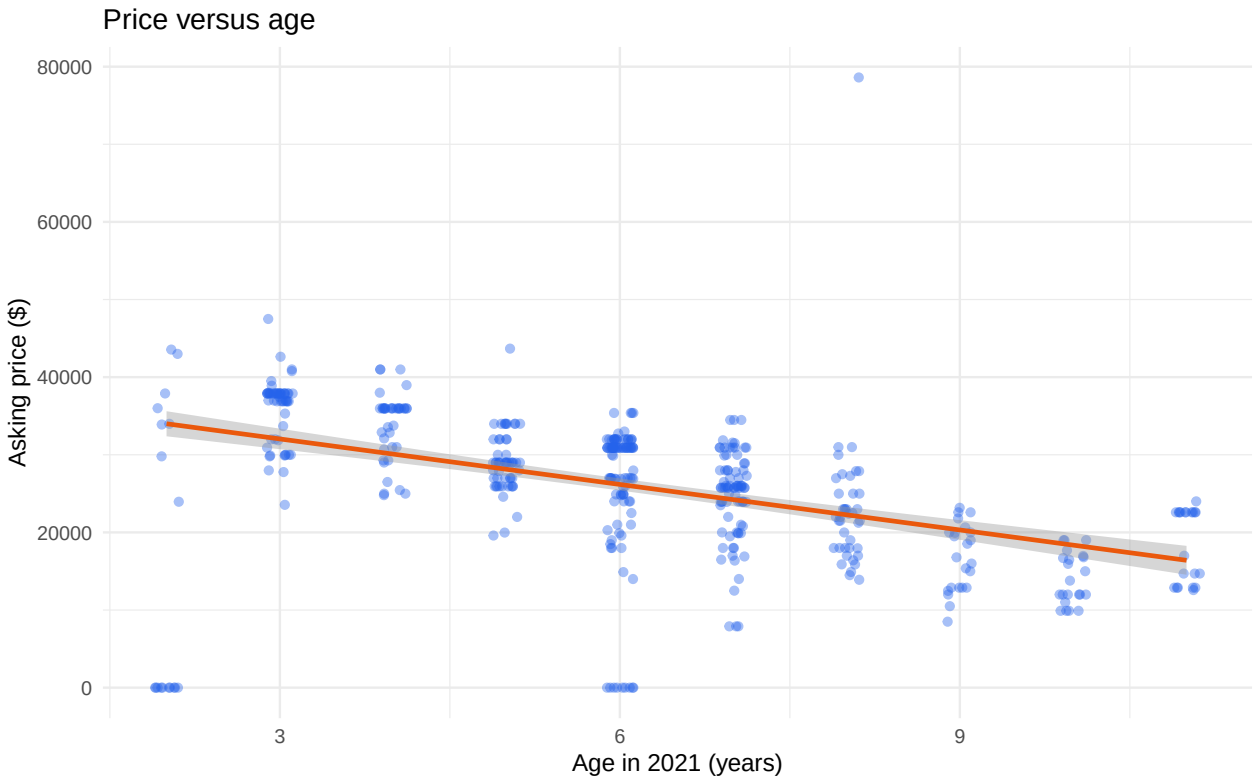
```
##      price            age            odometer
##  Min.   :    1   Min.   : 2.000   Min.   :     1
##  1st Qu.:20800   1st Qu.: 5.000   1st Qu.: 67787
##  Median :26995   Median : 6.000   Median :104385
##  Mean   :25980   Mean   : 6.102   Mean   : 98172
```

```
##  3rd Qu.:31990   3rd Qu.: 7.000   3rd Qu.:125000
##  Max.   :78609   Max.   :11.000   Max.   :254006
```

Exploratory data analysis

```
ggplot(vehicles, aes(odometer, price)) + geom_point(alpha=.4, color="#2563eb") + geom_smooth(method="lm
ggplot(vehicles, aes(age, price)) + geom_jitter(width=.12, alpha=.4, color="#2563eb") + geom_smooth(met
```





Baseline and nonlinear regression

```
mileage <- lm(price ~ odometer, data=vehicles)
age <- lm(price ~ age, data=vehicles)
additive <- lm(price ~ age + odometer, data=vehicles)
quadratic <- lm(price ~ age + I(age^2), data=vehicles)
second_order <- lm(price ~ age + odometer + I(age^2) + I(odometer^2) + age:odometer, data=vehicles)
model_list <- list("Mileage only"=mileage, "Age only"=age, "Age + mileage"=additive, "Quadratic age"=qua
comparison <- bind_rows(lapply(names(model_list), function(nm) {
 s <- glance(model_list[[nm]])
 data.frame(model=nm, r_squared=s$r.squared, adjusted_r_squared=s$adj.r.squared, AIC=s$AIC)
}))
knitr::kable(comparison, digits=3)
```

model	r_squared	adjusted_r_squared	AIC
Mileage only	0.210	0.208	9655.405
Age only	0.212	0.211	9653.826
Age + mileage	0.277	0.273	9616.593
Quadratic age	0.229	0.226	9645.826
Complete second order	0.310	0.302	9600.990

```
summary(additive)
```

```
##
## Call:
## lm(formula = price ~ age + odometer, data = vehicles)
##
```

```
## Residuals:
##    Min     1Q Median     3Q    Max
## -32296  -2529   1282   4116  53940
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.956e+04  1.127e+03  35.116  < 2e-16 ***
## age         -1.290e+03  1.981e+02  -6.511 1.96e-10 ***
## odometer    -5.817e-02  9.121e-03  -6.378 4.38e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8150 on 458 degrees of freedom
## Multiple R-squared:  0.2766, Adjusted R-squared:  0.2735
## F-statistic: 87.58 on 2 and 458 DF,  p-value: < 2.2e-16
```

A model with a higher R-squared is not automatically preferable: additional terms can overfit. Use the adjusted R-squared, nested tests for genuinely nested models, and holdout error to assess whether complexity helps.

```
anova(mileage, additive)
```

```
## Analysis of Variance Table
##
## Model 1: price ~ odometer
## Model 2: price ~ age + odometer
##   Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
## 1    459 3.3239e+10
## 2    458 3.0423e+10  1 2816114764 42.395 1.964e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(additive, second_order)
```

```
## Analysis of Variance Table
##
## Model 1: price ~ age + odometer
## Model 2: price ~ age + odometer + I(age^2) + I(odometer^2) + age:odometer
##   Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
## 1    458 3.0423e+10
## 2    455 2.9030e+10  3 1392766746 7.2765 8.925e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

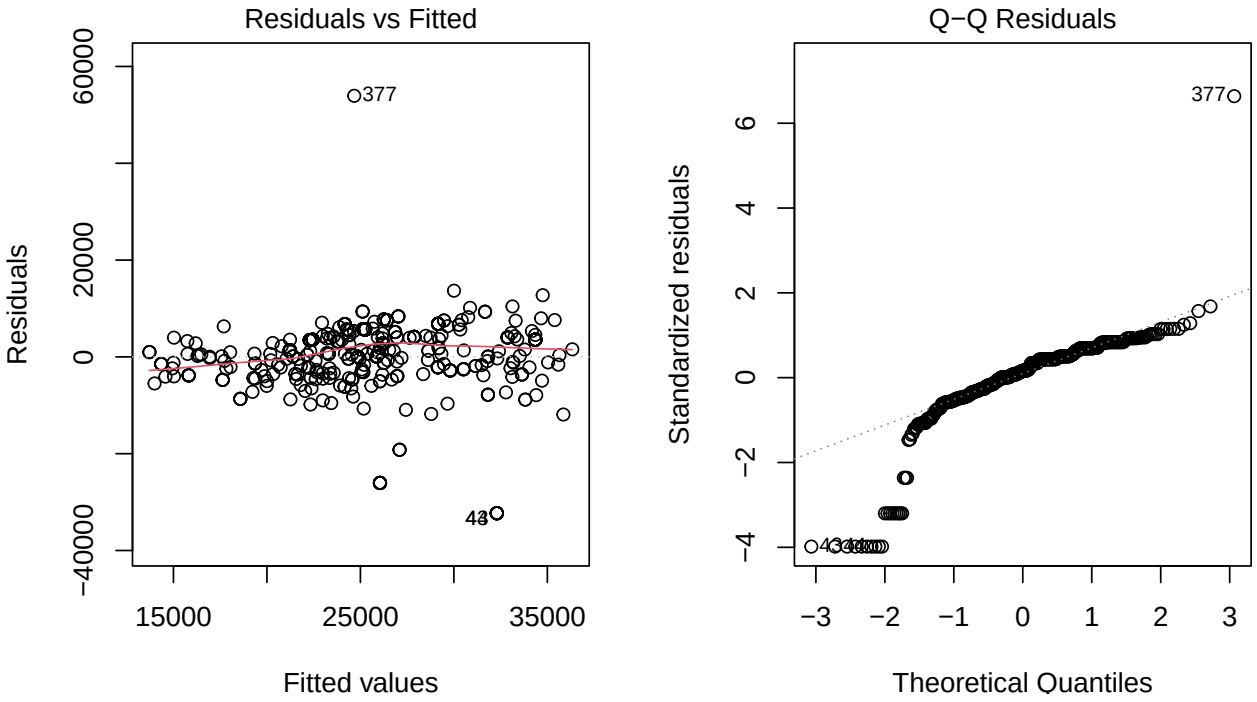
```
anova(quadratic, second_order)
```

```
## Analysis of Variance Table
##
## Model 1: price ~ age + I(age^2)
## Model 2: price ~ age + odometer + I(age^2) + I(odometer^2) + age:odometer
##   Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
## 1    458 3.2414e+10
## 2    455 2.9030e+10  3 3384438409 17.682 7.139e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual diagnostics

```
par(mfrow=c(1,2))
plot(additive, which=1)
plot(additive, which=2)
par(mfrow=c(1,1))
diagnostics <- data.frame(standardized_residual=rstandard(additive), leverage=hatvalues(additive), cooks
head(diagnostics[order(-diagnostics$cooks_distance),], 5)
```

```
##     standardized_residual    leverage cooks_distance
## 377              6.637658 0.005839727     0.08626698
## 43              -3.983474 0.010434444     0.05577344
## 44              -3.983474 0.010434444     0.05577344
## 50              -3.983474 0.010434444     0.05577344
## 67              -3.983474 0.010434444     0.05577344
```



Residual patterns may indicate nonconstant variance, nonnormal errors, or outliers. The table identifies observations worth inspecting rather than automatically deleting them.

Prediction intervals

```
new_listing <- data.frame(age=5, odometer=60000)
predict(additive, newdata=new_listing, interval="confidence", level=.95)
```

```
##        fit      lwr      upr
## 1 29622.06 28673.38 30570.74
```

```
predict(additive, newdata=new_listing, interval="prediction", level=.95)
```

```
##        fit      lwr      upr
## 1 29622.06 13577.64 45666.49
```

The confidence interval estimates the **mean** listing price at the specified age and mileage; the prediction interval describes a **single** new listing and is wider.

Holdout validation

```
set.seed(42)
train_idx <- sample(seq_len(nrow(vehicles)), size=floor(.8*nrow(vehicles)))
train <- vehicles[train_idx,]
test <- vehicles[-train_idx,]
formulas <- list("Mileage only"=price~odometer, "Age only"=price~age, "Age + mileage"=price~age+odometer
metrics <- bind_rows(lapply(names(formulas), function(nm) {
 fit <- lm(formulas[[nm]], data=train)
 predicted <- predict(fit, newdata=test)
 actual <- test$price
 data.frame(model=nm, MAE=mean(abs(actual-predicted)), RMSE=sqrt(mean((actual-predicted)^2)), test_R2=1-
}))
knitr::kable(metrics, digits=2)
```

model	MAE	RMSE	test_R2
Mileage only	5987.81	7993.43	0.21
Age only	5266.63	8112.26	0.18
Age + mileage	5103.41	7774.16	0.25
Quadratic age	5138.39	7856.29	0.23
Complete second order	5006.79	7328.79	0.33

The holdout split is random, not chronological. Results measure performance on a comparable sample of listings, not future market conditions.

Limitations and next steps

- Listings are not completed sales; listing quality and market conditions are unobserved.
- Age and mileage may be correlated; inspect coefficient stability and collinearity.
- Selection of a single model and year range limits generalizability.
- Improve the study by incorporating trim, condition, location, and listing date; validate on a later time period.

Reproducibility

Run `source("requirements.R")` once, then `rmarkdown::render("analysis.Rmd")` from the project directory. Data and plots are included in this repository.